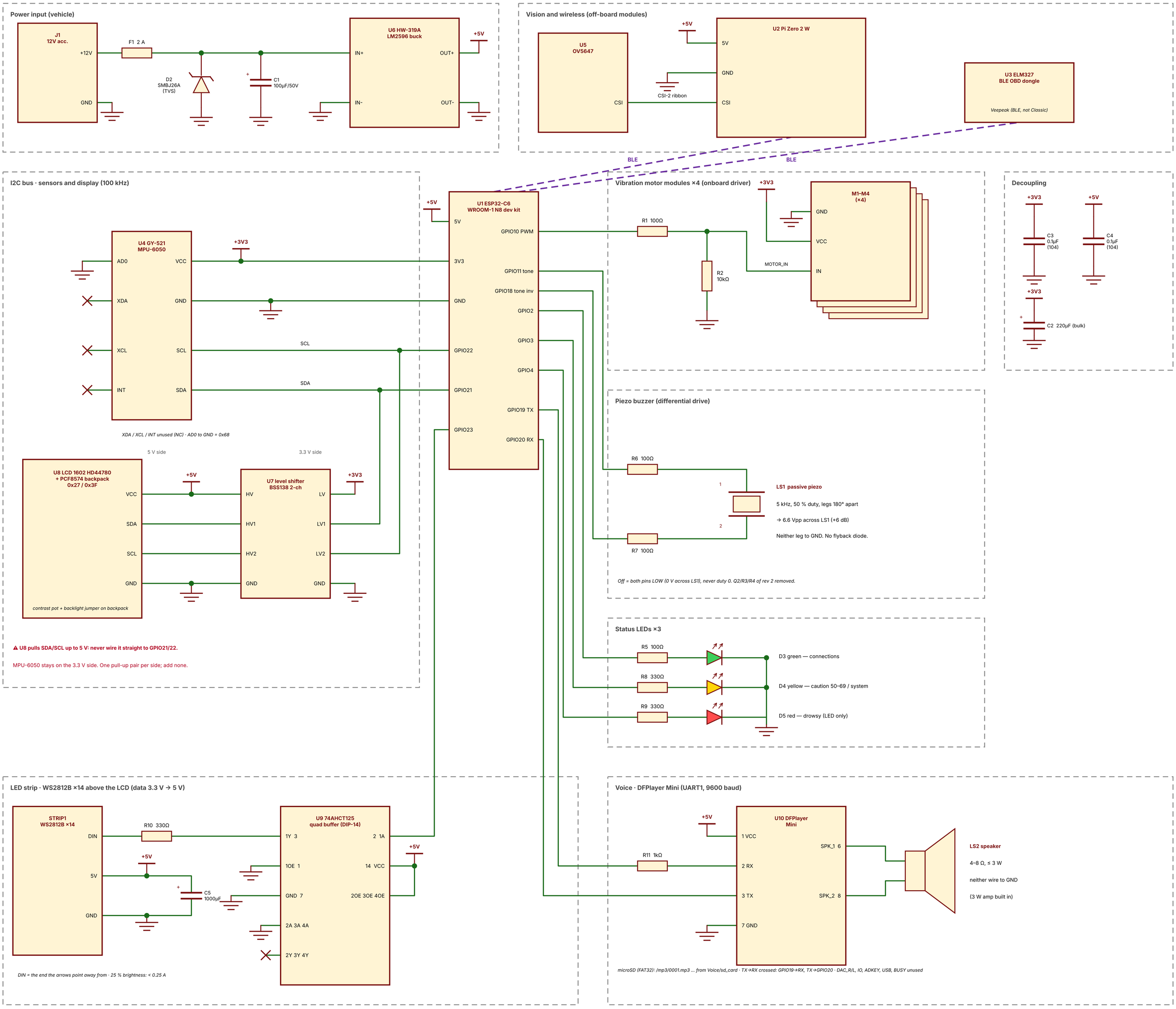
DRIFTS — Schematic (NewBrain architecture) · rev 5

Team 1 · ESP32-C6 dev kit + Raspberry Pi Zero 2 W · rev 5: WS2812B strip ×14 (GPIO23 + 74AHCT125), DFPlayer Mini voice (GPIO19/20) · rev 4: differential piezo · rev 3: status LEDs, I2C LCD + level shifter · generated drawing (not a KiCad file)



NOTES

1. U6 = HW-319A-V7.0.1 LM2596 module. Trim the output to 5.0 V with the trimmer BEFORE connecting the Pi or C6. Bench/demo: skip the 12 V front end and use a USB power bank.
2. Motor modules M1–M4 (GND, VCC, IN) have an onboard driver. IN ← GPIO10 (20 kHz PWM, 8-bit) via R1 100Ω; R2 10k holds IN low during reset.
3. Module VCC drawn from +3V3. Verify the module rating and that the C6 LDO can supply 4 × 60–100 mA plus the IMU; +5V is the alternative.
4. U4 GY-521: AD0→GND = 0x68. Wire.begin(21, 22). Breakout has pull-ups.
5. I2C runs at 100 kHz (PCF8574 limit), shared by U4 and U8. U7 BSS138 shifter: LV = +3V3, HV = +5V. U8 backpack is a 5 V part and pulls SDA/SCL to 5 V: never connect it directly to the C6. U8 address 0x27 (PCF8574) or 0x3F (PCF8574A), auto-detected.
6. LEDs: D3 green GPIO2 R5 (100Ω true green, 330Ω yellow-green), D4 yellow GPIO3 R8, D5 red GPIO4 R9. GPIO4 is a strapping pin: an LED to GND only, nothing else. Mount LEDs and LCD in the passenger's view.
7. LS1 passive piezo, differential: GPIO11→R6→LS1←R7←GPIO18, one LEDC channel, GPIO18 inverted; 5 kHz 50 %, 6.6 Vpp. Off = both pins LOW.
8. C3/C4: 0.1µF ceramic at +3V3 and +5V near U1; one per module VCC.
9. Wireless (dashed purple): Pi↔C6 and OBD dongle↔C6 over BLE. U3 must be BLE, not Classic. Camera U5 connects to the Pi only.
10. Do not use: 5, 8, 9, 15 (strapping); 6, 7 (JTAG); 12, 13 (USB); 16, 17 (UART0 to CH343); 14, 24–30 (flash). GPIO4 = red LED only.
11. Same-named flags (+5V, +3V3, GND) are one net, as in KiCad.
12. Firmware v18: LCD INTERVENTION 1/2 (score 50–69 / <50). LEDSTRIP, RAINBOW, SAY:n, VOICE test the rev 5 parts. See Hardware/pinout.md.
13. U9 74AHCT125 (VCC 5 V): the strip needs ≥ 3.5 V for a 1; the BSS138 (U7) is too slow for 800 kHz. Tie unused 2A–4A to GND, 2OE–4OE to 5 V.
14. U10 DFPlayer: 1k (R11) in series with its RX. Strip + DFPlayer share the 5 V rail: from the buck (U6) in the car; USB is fine on the bench.